O_2 releases depends on the need for O_2 in the O_2 in the surrounding tissues, so hemoglobin rarely if ever leaves all its O_2 behind → in capillaries CO_2 enters the bloodstream About 76% dissolves in the plasma, some of it remaining as dissolved CO_2 → the remainder forming bicarbonate.

About 23-24% of its ~~binding~~ binds to the amminо acids in the hemoglobin → forming a molecule known as carbaminohemoglobin. From the ~~capp~~ capillaries → the hemoglobin carries carbon dioxide back to the lungs. → where it releases it for exchange ~~of~~ of O_2.

Changes in the levels of RBCs can have significant effects on body's ability to effectively deliver oxygen to the tissues. Ineffective hematopoiesis results in insufficient numbers of RBCs & results in one of several forms of anemia. An overproduction of RBCs produces a condition ~~—~~ called Polycythemia. The ~~prim~~ primary drawback with polycythemia is not a failure to directly deliver enough O_2 to the tissues, but rather the increased viscosity of the blood → which makes it more difficult for the heart to circulate the blood.

In patients with insufficient hemoglobin, the tissues may not receive sufficient O_2, resulting in another form of anemia. In determining O_2 of tissues, the value of greatest interest in healthcare is the % saturation i.e. of hemoglobin sites occupied by O_2, patient blood. { Percent sat }